Who Pays to Compete? Wealth, a Fixed Fee and Entry by Rank

Proofs addendum

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The results below are proved in Lean and are more general than the statements of `MANUSCRIPT.tex`. Notation is that of Appendix A there.

Dominance with $\mu \le 1$. $F \le G$ holds for every $\mu \le 1$, with no lower bound on μ, because the term μr is common to both scores.
Lean: `EntryContestP1P2.p2_fosd`.

Saturation for every incumbent. $\Delta(m) \le V/(m+1)$ for every Q needs no atoms in the investor's law only. The non-investor's law and the incumbent's law C can be any probability laws.
Lean: `EntryContestP6P7.p7_Delta_le`, `EntryContestP6P7.p7_cap`.

The benchmark limit from both sides. $\Delta(m) \to V/(m+2)$ as $\mu \to 0$ in a two-sided neighbourhood of 0, for any law of r, when $s \ge 0$ has no atoms.
Lean: `EntryContestP6P7.p6_limit`.

Strictness under a spread with weaker hypotheses. The strict fall of k^* under a strong enough linear spread needs only a non-increasing κ above c, the divergence of κ at c and a non-increasing Δ. Neither the continuity of κ nor the strictness of burden-monotonicity is used.
Lean: `EntryContestKappa.p9_strict`, `EntryContestKappa.p9_monotone`.

The divergence primitive. For u continuous at c, $\kappa(w) \to \infty$ as $w \downarrow c$ exactly when $u(x) \to -\infty$ as $x \downarrow 0$.
Lean: `EntryContestKappa.kappa_diverges_iff`.

Burden-monotonicity without curvature. If $u'(w) < u'(w-c)$ for every $w > c$, κ is strictly decreasing on (c, ∞). A convex stretch of u wider than c rules burden-monotonicity out, and every width below c is attained by an admissible u.
Lean: `EntryContestBM.burden_strictAnti_of_deriv`, `EntryContestBM.not_burden_of_convex_stretch`, `EntryContestBM.admissible_width`.

The no-collapse bound in general. When non-investor scores never exceed x_G, for $m < Q$

$$\Delta(m,Q) \ge V\Big(\int_{x \ge x_G} CF^m \, dF - \frac{1}{Q-m}\Big).$$

Lean: `EntryContestRefute.r1_lower_bound`.

Count invariance over all indices. The count-invariance argument holds with the outsider condition imposed at every index, not only below Q.
Lean: `EntryContestMM.count_invariance`.